2200 KV Brushless DC (BLDC) Motor

Component Specification Sheet ? CBU Gyroscope Project 2026

Electrical Specifications

KV Rating	2200 KV (RPM per volt)
Maximum continuous current	10 A
Peak current (short-term)	12 A
Recommended voltage range	2S-3S LiPo (7.4 V - 11.1 V)
Project operating voltage	approx. 4.84 V
Winding resistance (R)	0.12 ohm
Back-EMF constant (ke)	0.00434 V.s/rad

Performance at Project Operating Point (8000 RPM)

Operating speed	8,000 RPM
Required voltage	4.84 V
Current draw	10.0 A
Electrical input power	48.4 W
Mechanical power (stored in flywheel)	1,756.7 W
Efficiency (approx.)	>85%

Physical Specifications

Motor type	Outrunner / Inrunner Brushless DC (BLDC)
Number of poles	typically 12 (varies by model)
Shaft diameter	3.17 mm (1/8 inch)
Mounting hole pattern	16 mm x 19 mm (standard)
Mass (approx.)	50-70 g

Control Interface

Control signal type	PWM (servo-standard)
PWM frequency	50 Hz
Pulse width range	1000 us (stop) to 2000 us (full throttle)
PWM controller used	ESP32 (LEDC peripheral)
ESC type	Custom-built module (3-phase commutation)

Selection Rationale (Project-Specific)

Why 2200 KV?	Target speed achievable at safe, low voltage (~4.84 V)
Why BLDC?	No brush wear; suitable for 24-hour continuous operation
Starting torque limitation	Low; requires manual spin-up to ~1000 RPM
Coupling used	Rubber spider coupling for vibration isolation

Specification summary compiled for CBU Final Year Project 2026.